

DC Biasing – BJTs

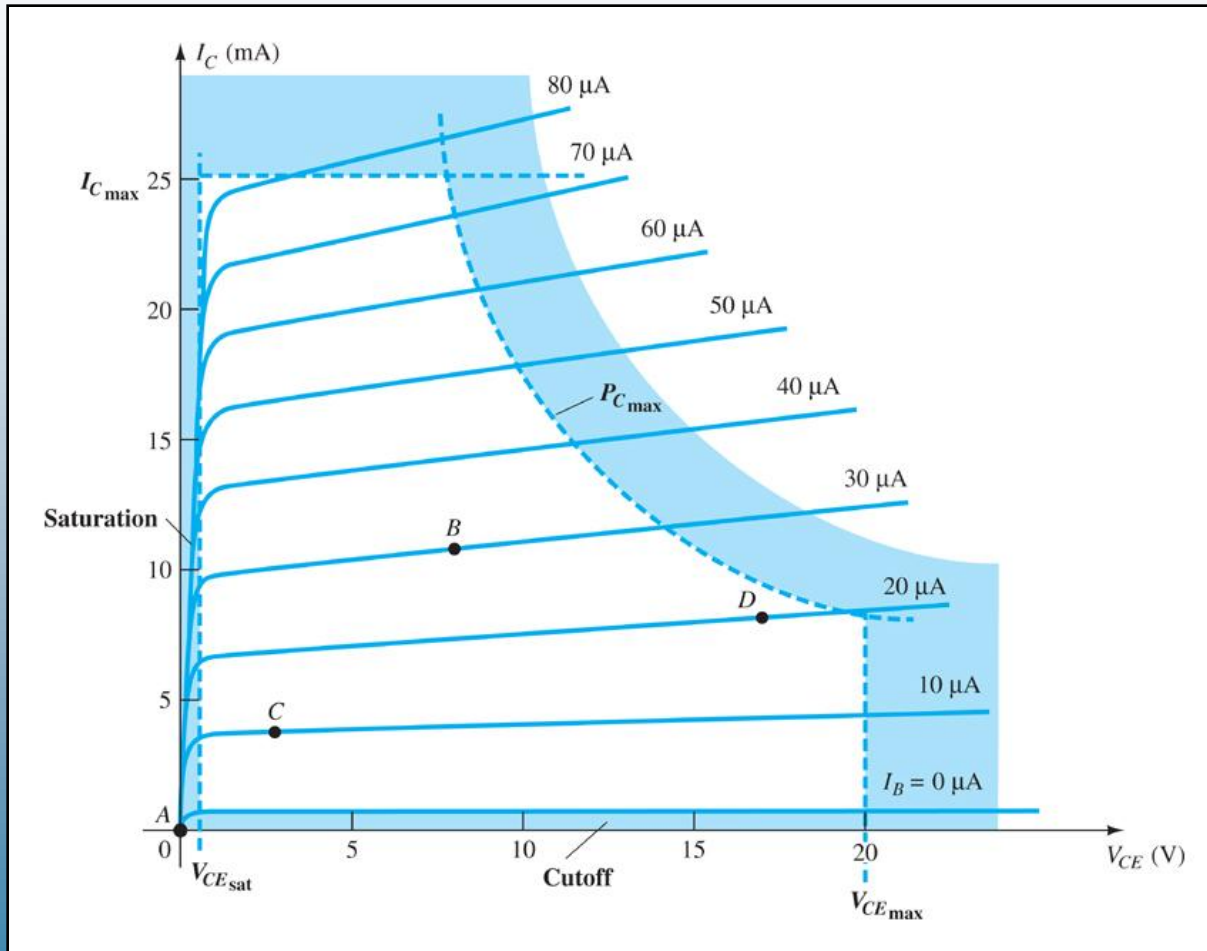
Topic 4 (Chapter 4)

(Some materials are from Malvino's book)

Biasing

Biasing: Applying DC voltages to a transistor in order to turn it on so that it can amplify AC signals.

4.2 Operating Point



The DC voltages establishes an operating or *quiescent point* called the **Q-point**.

The Three Operating Regions

Active or Linear Region Operation

- Base–Emitter junction is forward biased
- Base–Collector junction is reverse biased
- **BJT works as an Amplifier**

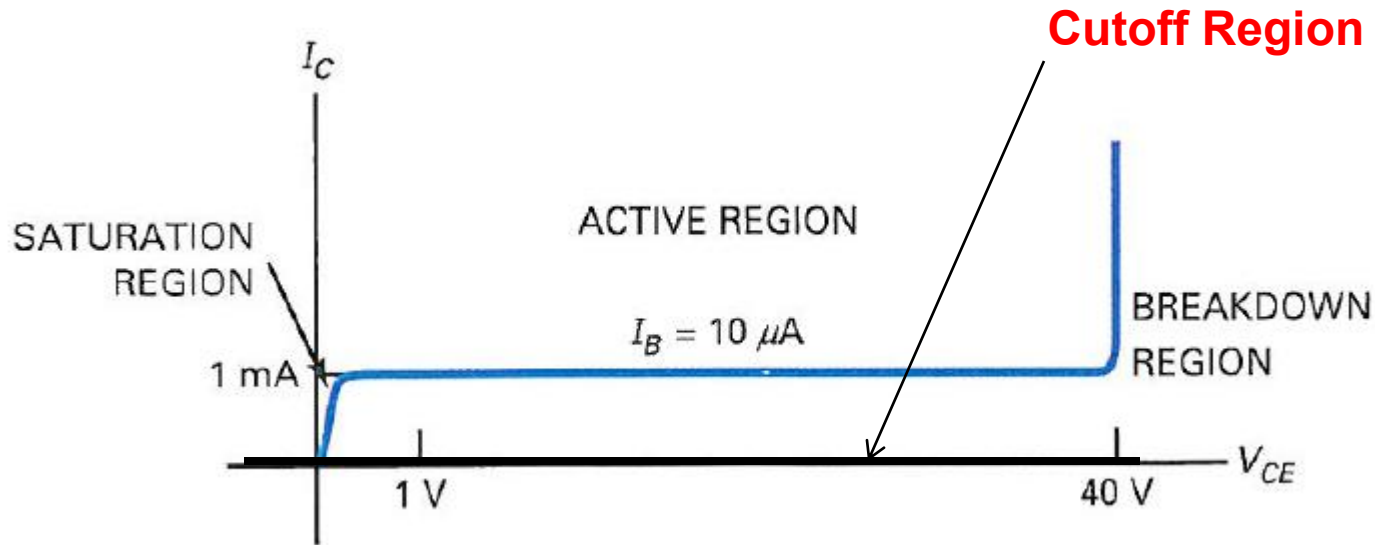
Cutoff Region Operation

- Base–Emitter junction is reverse biased
- Base–Collector junction is reverse biased
- **BJT works as an OFF Switch**

Saturation Region Operation

- Base–Emitter junction is forward biased
- Base–Collector junction is forward biased or near forward bias
- **BJT works as an ON Switch**

Regions of operation



1. **Active** - - - used for linear amplification
 2. **Cutoff** - - - used in switching applications (OFF)
 3. **Saturation** - - - used in switching applications (ON)
- **Breakdown** - - - can destroy the transistor and should be avoided

DC Biasing Circuits

Fixed-bias or Base-Bias

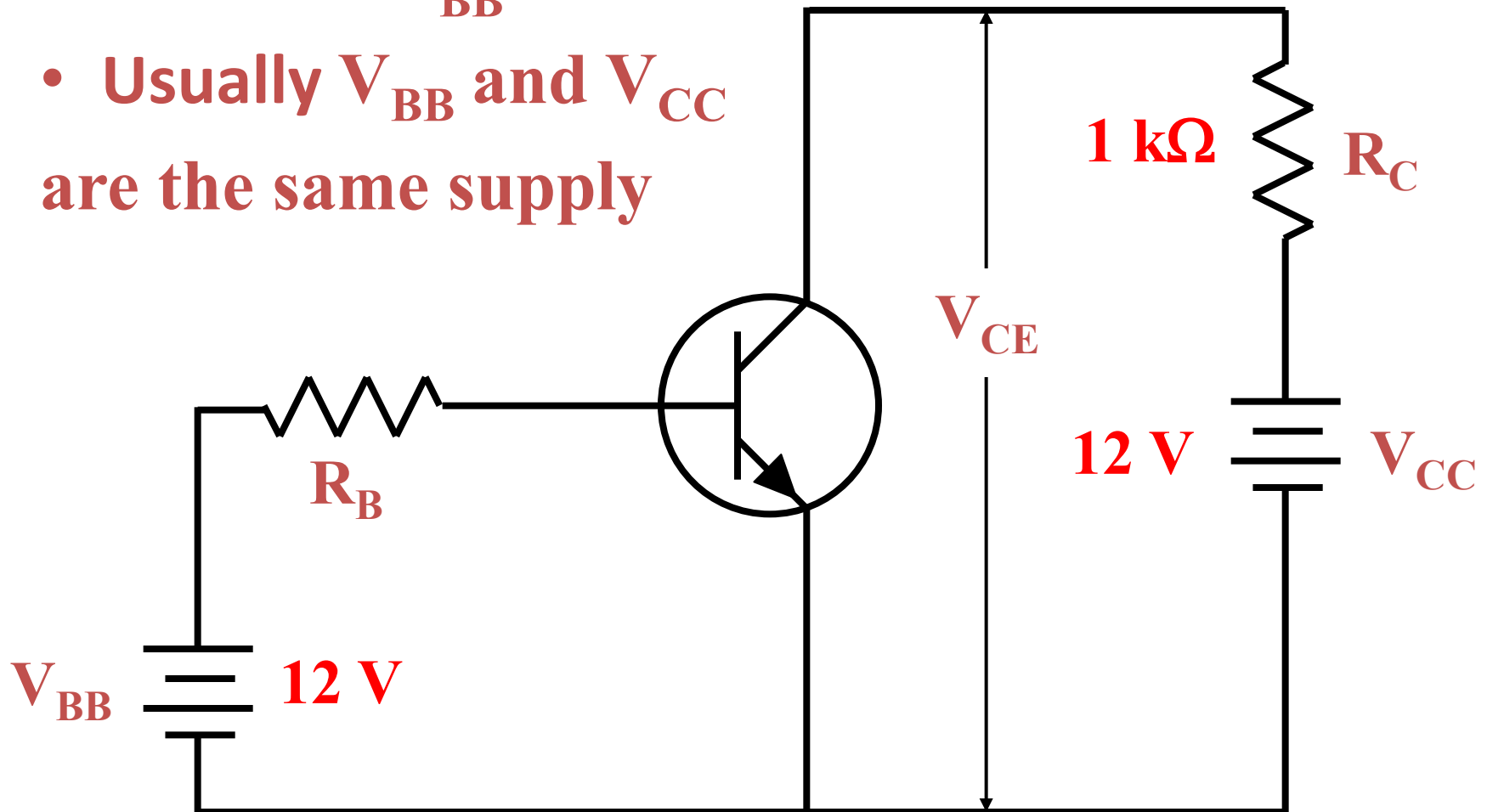
Emitter Bias

Voltage divider bias circuit

DC bias with voltage feedback

4.3 Base-Bias or Fixed-Bias

- Setting up a fixed value of **base** current
- Base current remains constant for given values of V_{BB}
- Usually V_{BB} and V_{CC} are the same supply



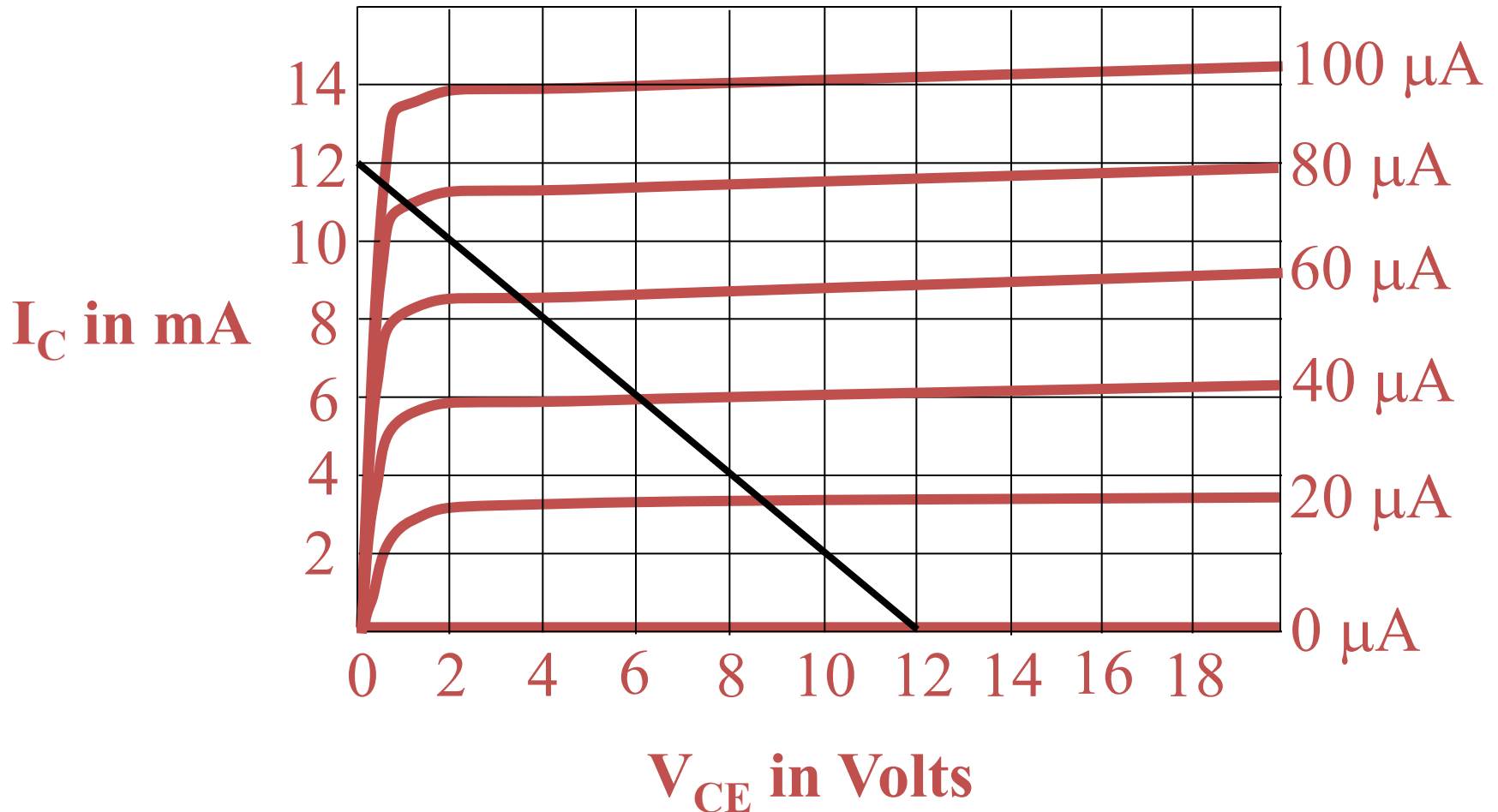
Load line

- A visual summary of all the possible transistor operating points
- Connects saturation current (I_{Csat}) to cutoff voltage ($V_{CEcutoff}$)

Load line

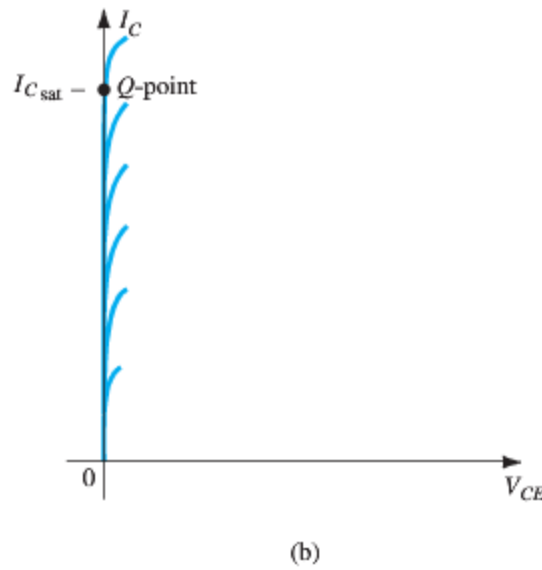
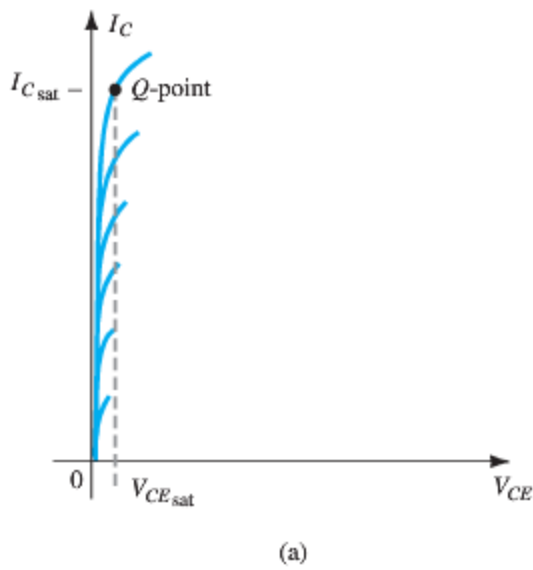
$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

A graph of this equation produces a load line.



Saturation

When the transistor is operating in **saturation**, current through the transistor is at its *maximum* possible value.

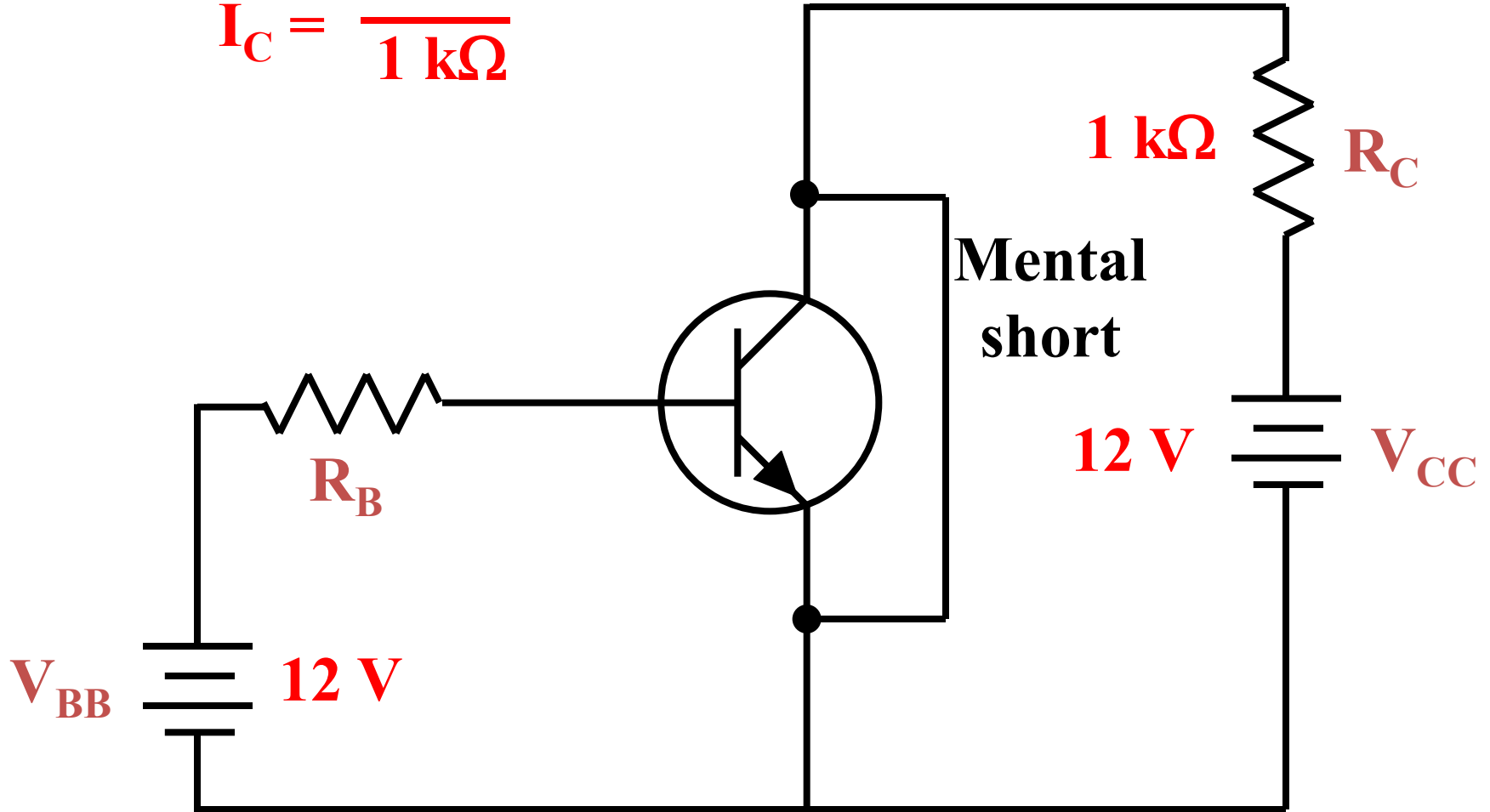


$$I_{Csat} = \frac{V_{CC}}{R_C}$$

$$V_{CE} \approx 0 \text{ V}$$

Understanding Saturation

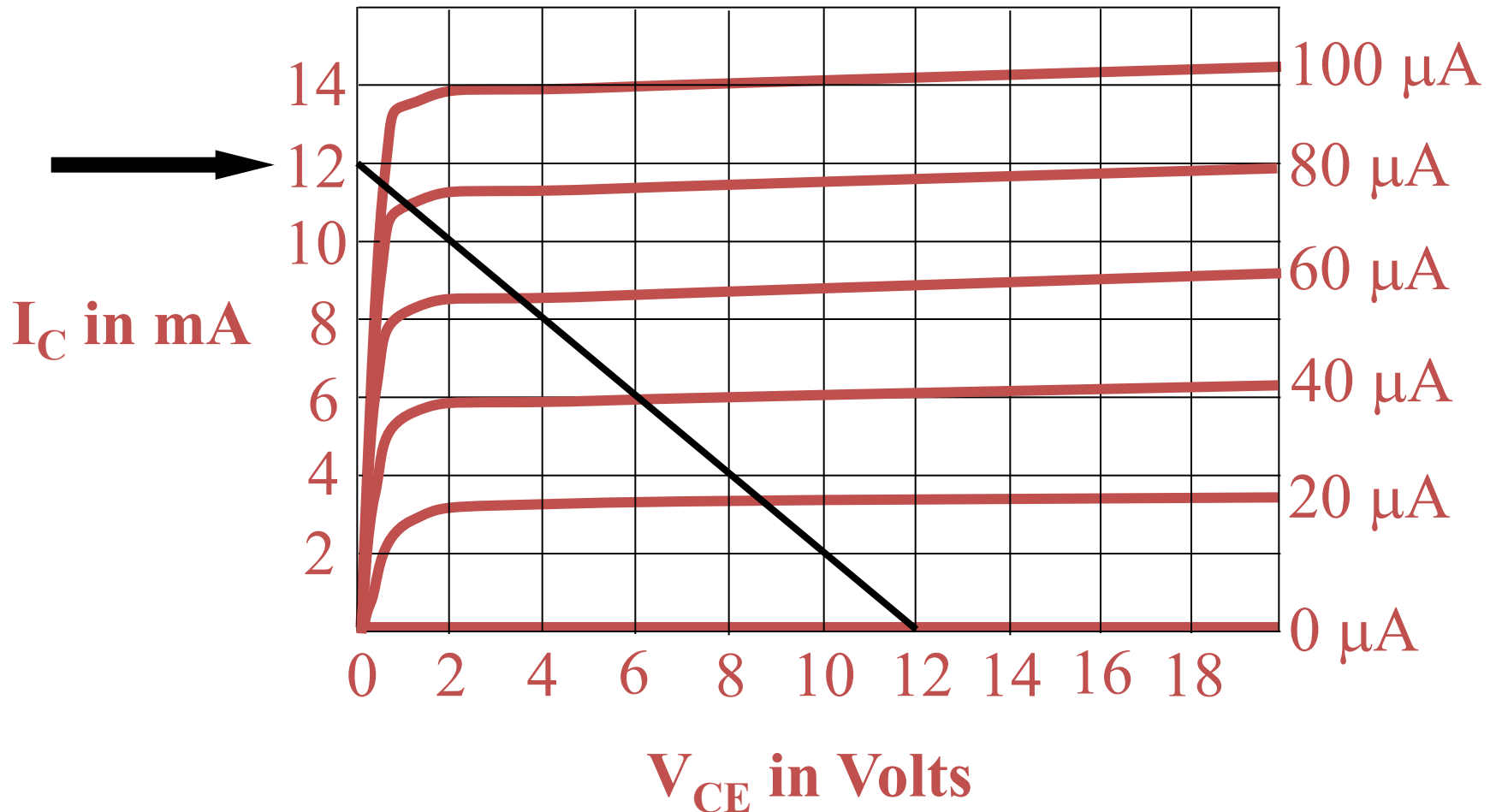
$$I_C = \frac{12\text{ V}}{1\text{ k}\Omega}$$



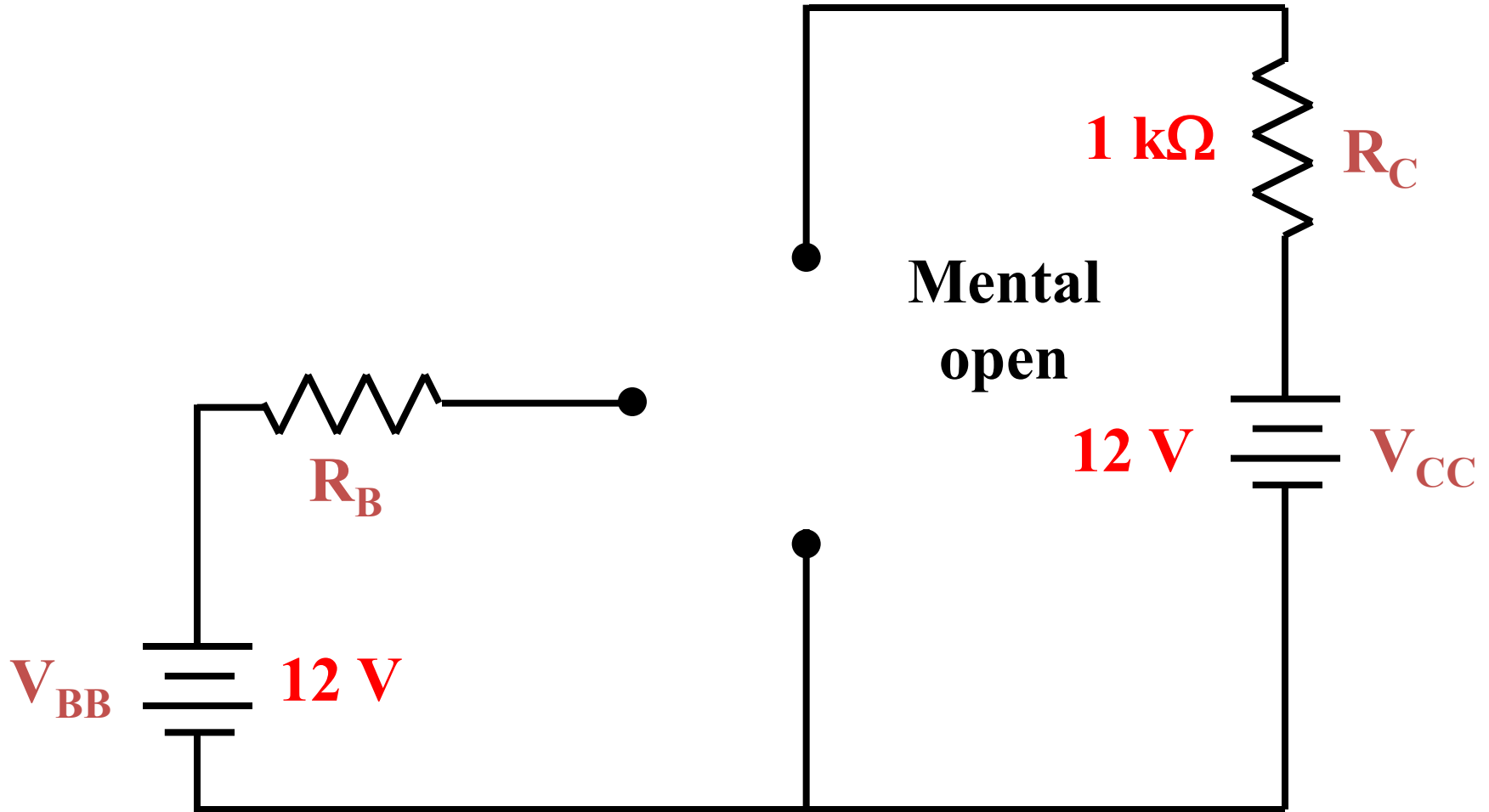
Understanding Saturation

$$I_C = \frac{12\text{ V}}{1\text{ k}\Omega} = 12\text{ mA}$$

This is the **Saturation (maximum) current.**

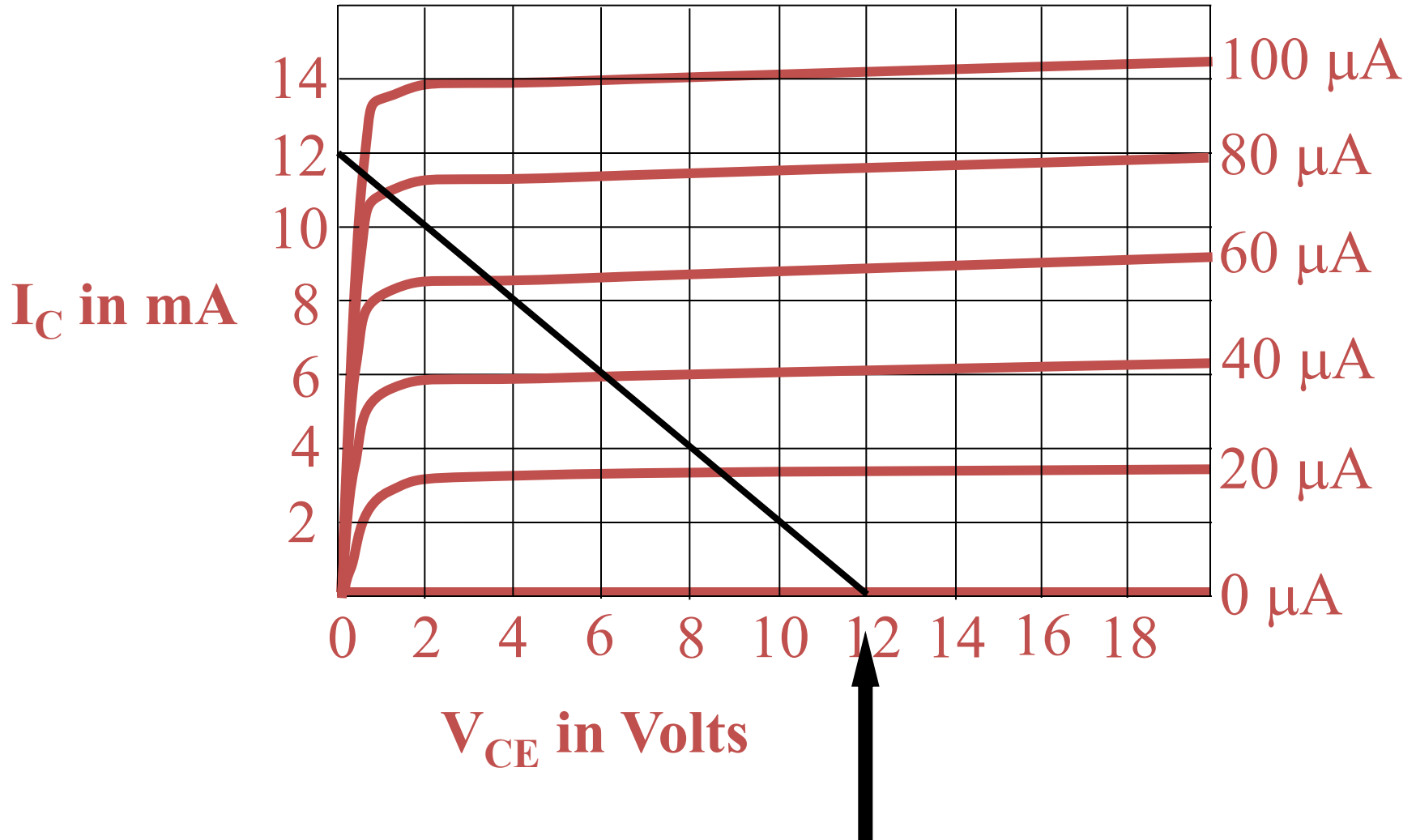


Understanding Cutoff



Understanding Cutoff

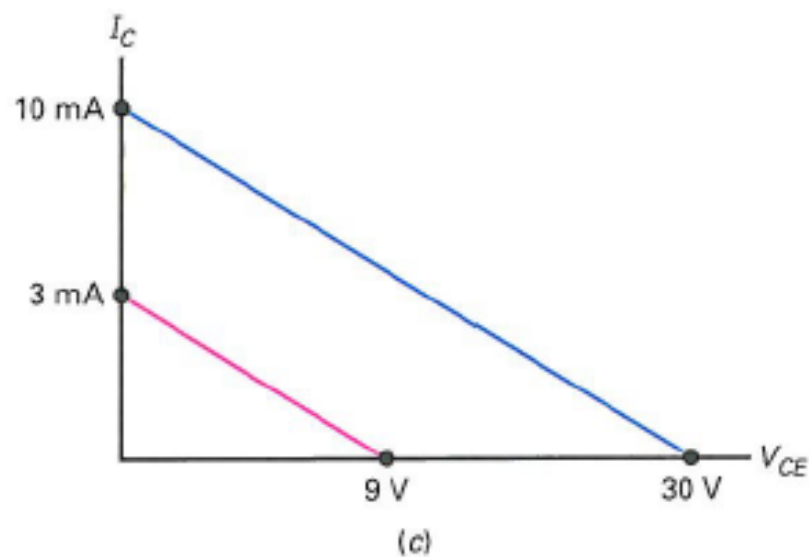
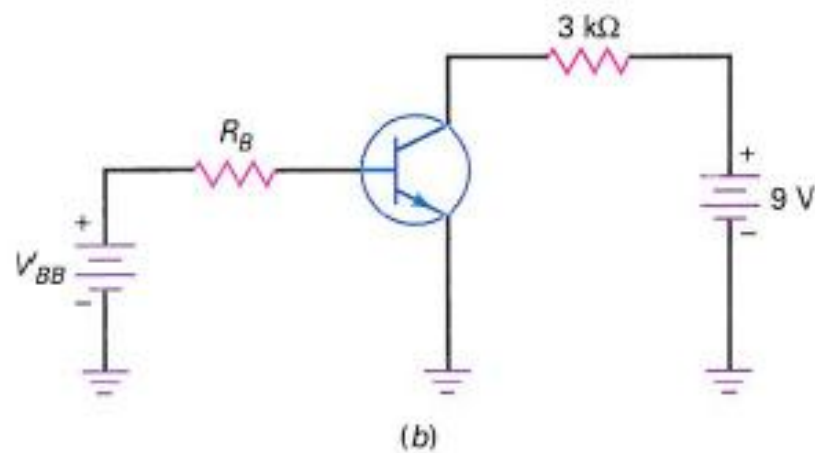
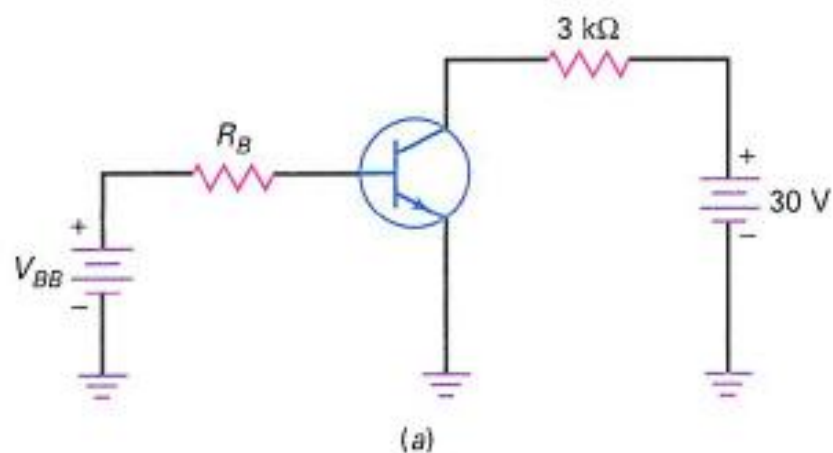
$$V_{CE(\text{cutoff})} = V_{CC}$$



Example 7-1

What are the saturation current and the cutoff voltage in Fig. 7-4a?

Calculate the saturation and cutoff values for Fig. 7-4b.

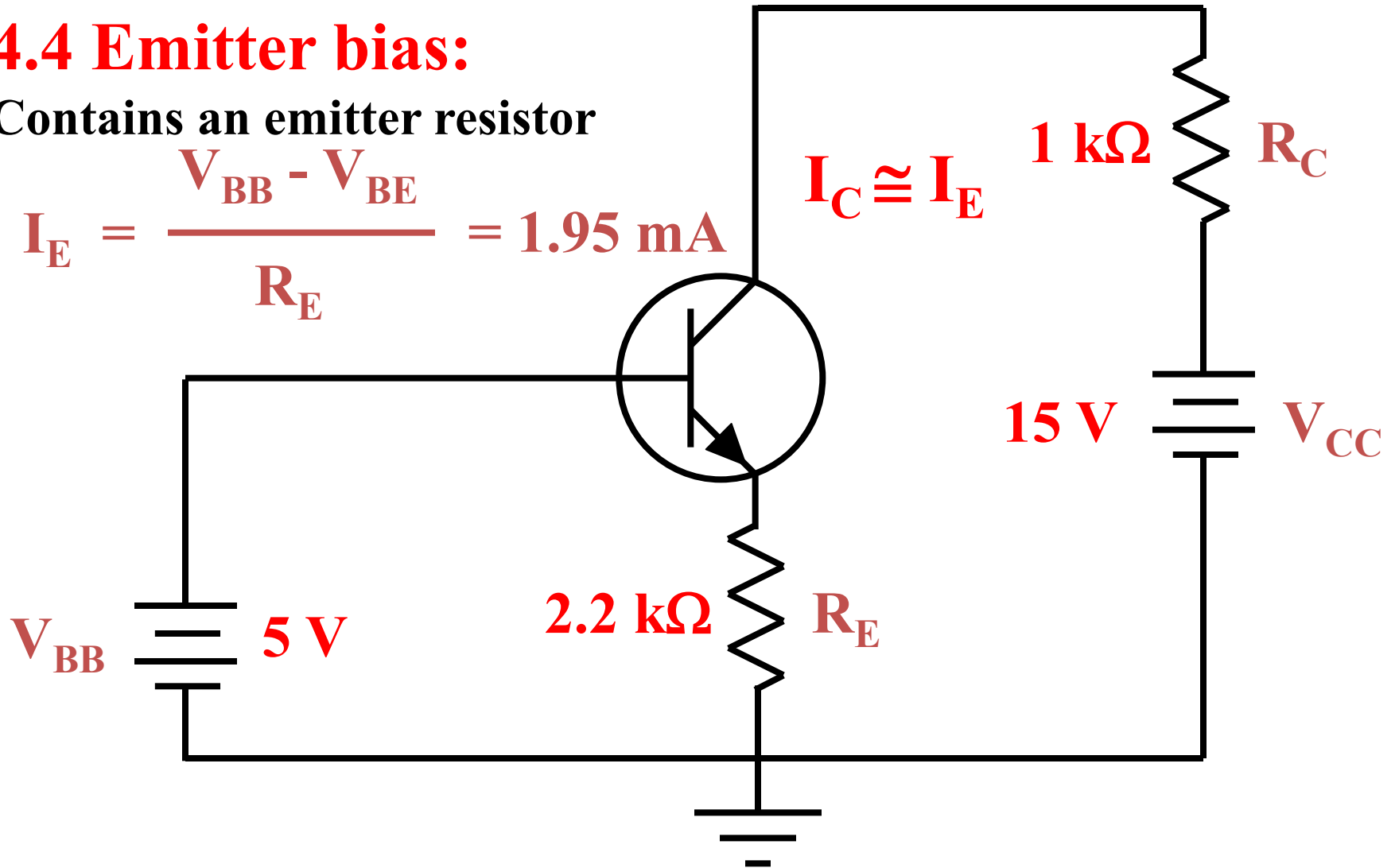


4.4 Emitter bias:

Contains an emitter resistor

$$I_E = \frac{V_{BB} - V_{BE}}{R_E} = 1.95 \text{ mA}$$

$$I_C \cong I_E \quad 1 \text{ k}\Omega \quad R_C$$



$$V_C = 15 \text{ V} - (1.95 \text{ mA})(1 \text{ k}\Omega) = 13.1 \text{ V}$$

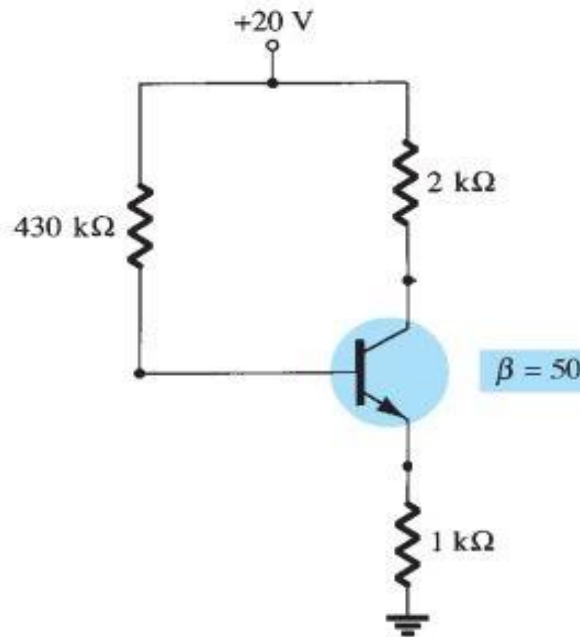
$$V_{CE} = 13.1 \text{ V} - 4.3 \text{ V} = 8.8 \text{ V}$$

BJT DC biasing

- First give appropriate directions of the three currents (I_E , I_B and I_C)
- Find α and β
- Apply KVL in the base-emitter loop. $V_{BE} = 0.7V$. Find I_B
- Find $I_C = \beta I_B$ and $I_E = I_B + I_C = (\beta + 1)I_B$ from I_B
- Apply KVL in the collector-emitter loop. Find V_{CE}
- Apply KVL in the base loop to find V_B
- $V_{BE} = 0.7V = V_B - V_E$. Find V_E
- $V_{CE} = V_C - V_E$. Find V_C
- Apply KVL in the emitter loop to find V_E

EXAMPLE 4.4 For the emitter-bias network of Fig. 4.23, determine

- a. I_B .
- b. I_C .
- c. V_{CE} .
- d. V_C .
- e. V_E .
- f. V_B .
- g. V_{BC} .



Solution:

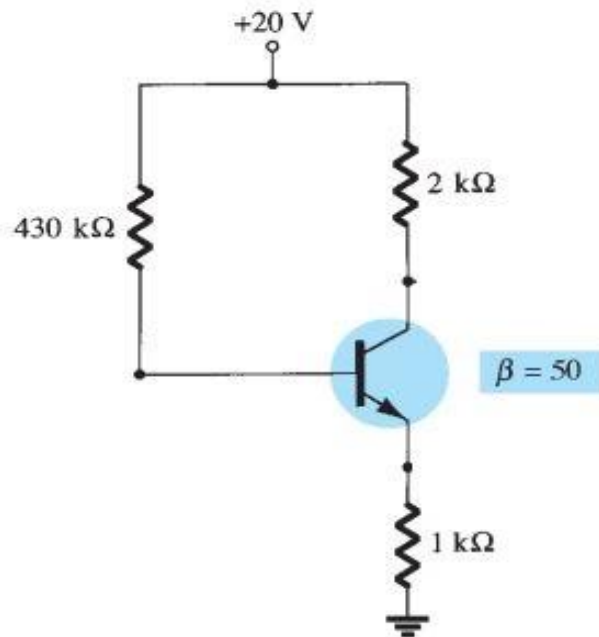
a. Eq. (4.17):
$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{430 \text{ k}\Omega + (51)(1 \text{ k}\Omega)}$$
$$= \frac{19.3 \text{ V}}{481 \text{ k}\Omega} = 40.1 \mu\text{A}$$

b.
$$I_C = \beta I_B$$
$$= (50)(40.1 \mu\text{A})$$
$$\cong 2.01 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$
$$= 20 \text{ V} - (2.01 \text{ mA})(2 \text{ k}\Omega + 1 \text{ k}\Omega) = 20 \text{ V} - 6.03 \text{ V}$$
$$= 13.97 \text{ V}$$

EXAMPLE 4.4 For the emitter-bias network of Fig. 4.23, determine

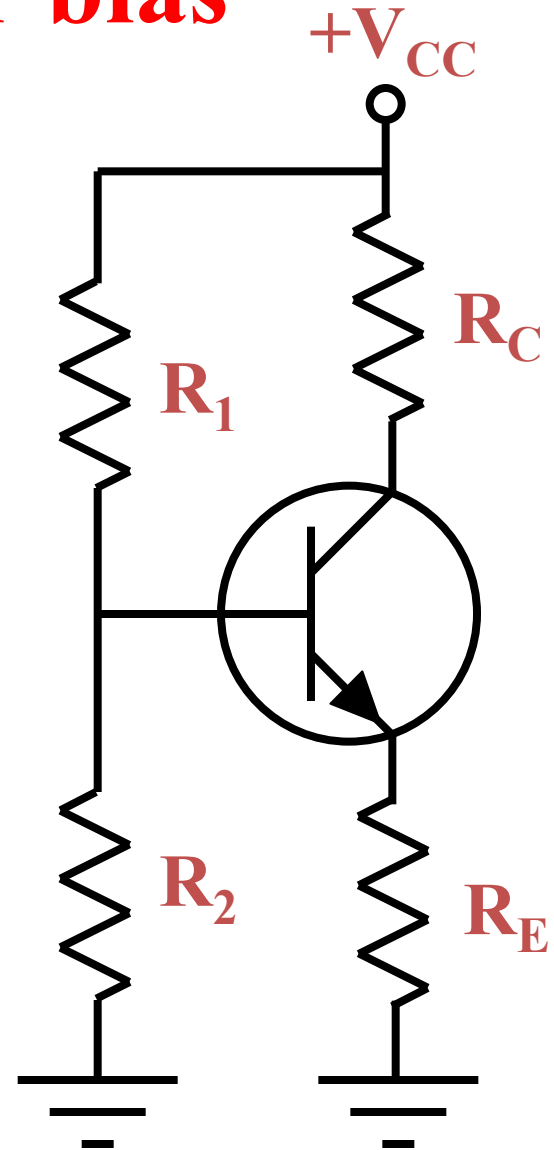
- a. I_B .
- b. I_C .
- c. V_{CE} .
- d. V_C .
- e. V_E .
- f. V_B .
- g. V_{BC} .



$$\begin{aligned}\text{d. } V_C &= V_{CC} - I_C R_C \\ &= 20 \text{ V} - (2.01 \text{ mA})(2 \text{ k}\Omega) = 20 \text{ V} - 4.02 \text{ V} \\ &= \mathbf{15.98 \text{ V}} \\ \text{e. } V_E &= V_C - V_{CE} \\ &= 15.98 \text{ V} - 13.97 \text{ V} \\ &= \mathbf{2.01 \text{ V}} \\ \text{or } V_E &= I_E R_E \cong I_C R_E \\ &= (2.01 \text{ mA})(1 \text{ k}\Omega) \\ &= \mathbf{2.01 \text{ V}} \\ \text{f. } V_B &= V_{BE} + V_E \\ &= 0.7 \text{ V} + 2.01 \text{ V} \\ &= \mathbf{2.71 \text{ V}} \\ \text{g. } V_{BC} &= V_B - V_C \\ &= 2.71 \text{ V} - 15.98 \text{ V} \\ &= \mathbf{-13.27 \text{ V}} \text{ (reverse-biased as required)}\end{aligned}$$

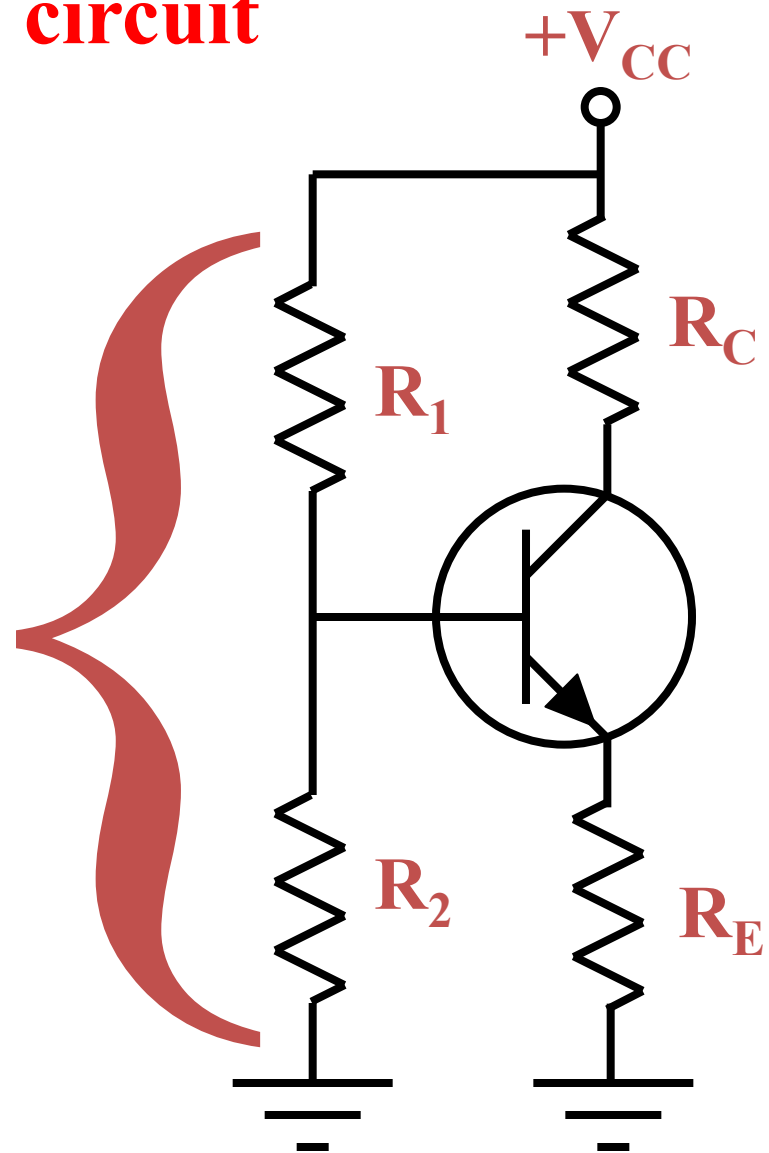
4.5 Voltage divider bias

- Base circuit contains a **voltage divider**
- Most widely used
- Known as **VDB**



Voltage divider bias circuit

R_1 and R_2 form
a voltage divider



Divider analysis:

$$R_{th} = R_1 \parallel R_2$$

$$E_{th} = \frac{R_2}{R_1 + R_2} V_{CC}$$

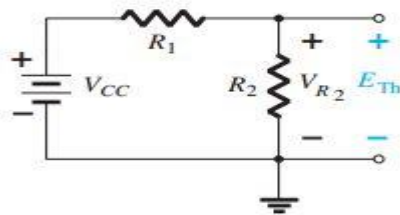


FIG. 4.33
Determining E_{Th} .

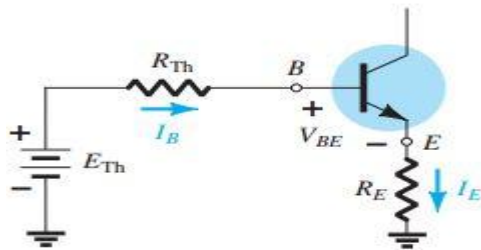


FIG. 4.34
Inserting the Thévenin equivalent circuit.

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$

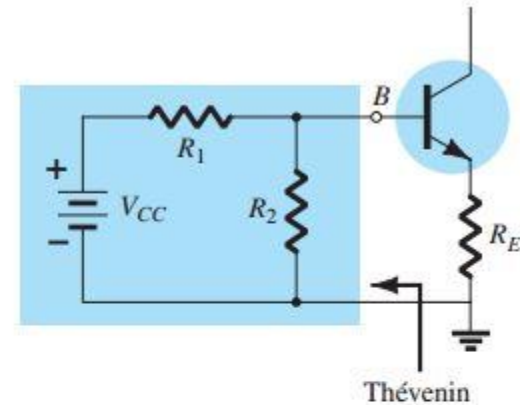


FIG. 4.31
Redrawing the input side of the network of Fig. 4.28.

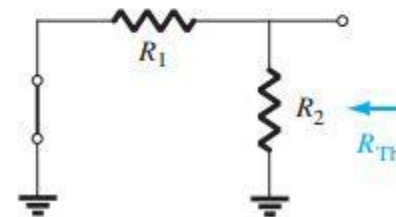


FIG. 4.32
Determining R_{Th} .

EXAMPLE 4.8 Determine the dc bias voltage V_{CE} and the current I_C for the voltage-divider configuration of Fig. 4.35.

Solution: Eq. (4.28): $R_{Th} = R_1 \parallel R_2$

$$= \frac{(39 \text{ k}\Omega)(3.9 \text{ k}\Omega)}{39 \text{ k}\Omega + 3.9 \text{ k}\Omega} = 3.55 \text{ k}\Omega$$

Eq. (4.29): $E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2}$

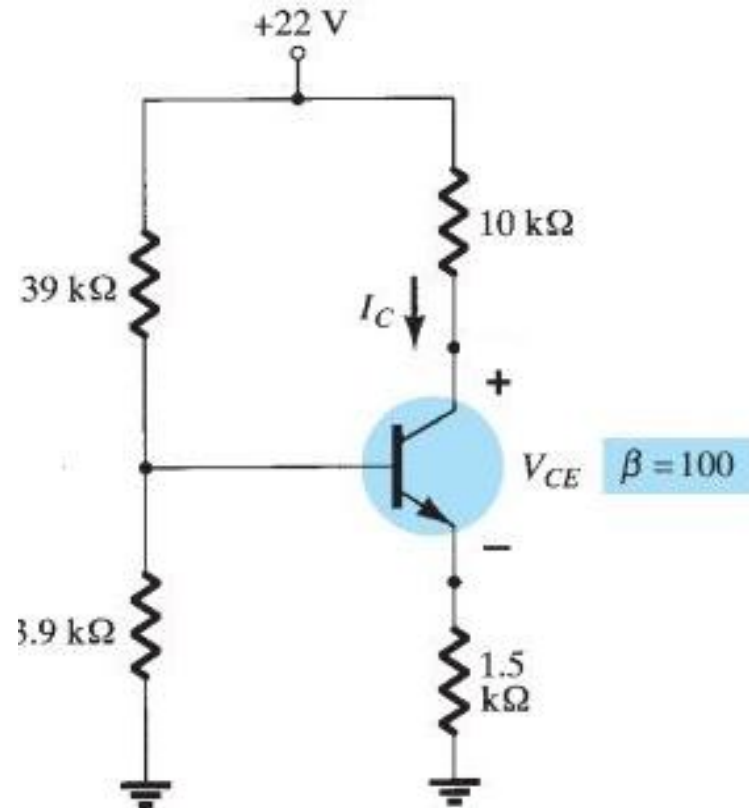
$$= \frac{(3.9 \text{ k}\Omega)(22 \text{ V})}{39 \text{ k}\Omega + 3.9 \text{ k}\Omega} = 2 \text{ V}$$

Eq. (4.30): $I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$

$$= \frac{2 \text{ V} - 0.7 \text{ V}}{3.55 \text{ k}\Omega + (101)(1.5 \text{ k}\Omega)} = \frac{1.3 \text{ V}}{3.55 \text{ k}\Omega + 151.5 \text{ k}\Omega}$$
$$= 8.38 \mu\text{A}$$

$$I_C = \beta I_B$$
$$= (100)(8.38 \mu\text{A})$$
$$= \mathbf{0.84 \text{ mA}}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$
$$= 22 \text{ V} - (0.84 \text{ mA})(10 \text{ k}\Omega + 1.5 \text{ k}\Omega)$$
$$= 22 \text{ V} - 9.66 \text{ V}$$
$$= \mathbf{12.34 \text{ V}}$$



4.6 Collector-feedback bias

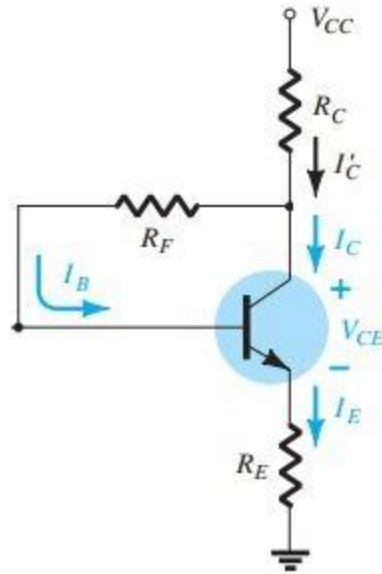


FIG. 4.38

DC bias circuit with voltage feedback.

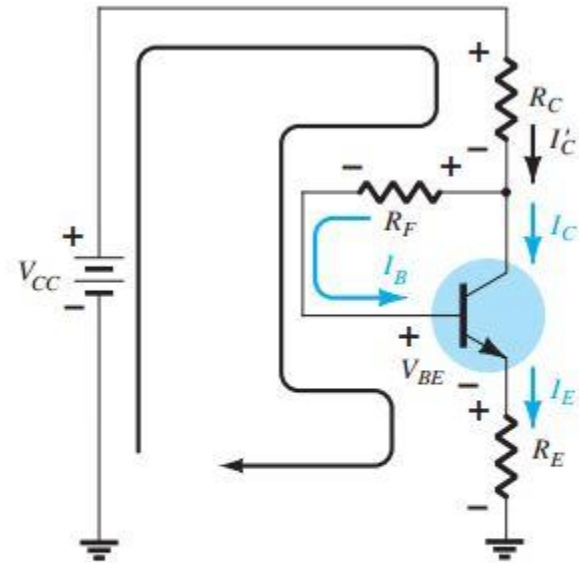


FIG. 4.39

Base-emitter loop for the network of Fig. 4.38.

Our text calls it **DC Bias With Voltage Feedback**

Current through $R_C = I_{R_C} = I_C'$

$$I_C' = I_C + I_B$$

EXAMPLE 4.12 Determine the quiescent levels of I_{CQ} and V_{CEQ} for the network of Fig. 4.41.

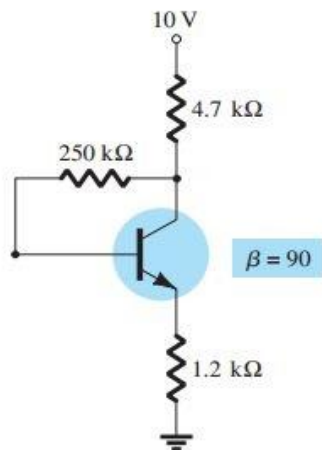


FIG. 4.41
Network for Example 4.12.

$$\begin{aligned}
 \therefore I_B &= \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)} \\
 &= \frac{10 \text{ V} - 0.7 \text{ V}}{250 \text{ k}\Omega + (90)(4.7 \text{ k}\Omega + 1.2 \text{ k}\Omega)} \\
 &= \frac{9.3 \text{ V}}{250 \text{ k}\Omega + 531 \text{ k}\Omega} = \frac{9.3 \text{ V}}{781 \text{ k}\Omega} \\
 &= 11.91 \mu\text{A} \\
 I_{CQ} &= \beta I_B = (90)(11.91 \mu\text{A}) \\
 &= \mathbf{1.07 \text{ mA}} \\
 V_{CEQ} &= V_{CC} - I_C(R_C + R_E) \\
 &= 10 \text{ V} - (1.07 \text{ mA})(4.7 \text{ k}\Omega + 1.2 \text{ k}\Omega) \\
 &= 10 \text{ V} - 6.31 \text{ V} \\
 &= \mathbf{3.69 \text{ V}}
 \end{aligned}$$